

Oligodendrocytes in Alzheimer's disease pathophysiology

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Our understanding of Alzheimer's disease (AD) has transformed from a purely neuronal perspective to one that acknowledges the involvement of glial cells. Despite remarkable progress in unraveling the biology of microglia, astrocytes and vascular elements, the exploration of oligodendrocytes in AD is still in its early stages. Contrary to the traditional notion of oligodendrocytes as passive bystanders in AD pathology, emerging evidence indicates their active participation in and reaction to amyloid and tau pathology. Oligodendrocytes undergo a functional transition to a disease-associated state, engaging in immune modulation, stress responses and cellular survival. Far from being inert players, they appear to serve a dual role in AD pathogenesis, potentially offering defense mechanisms against pathology while also contributing to disease progression. This Review explores recent advancements in understanding the roles of oligodendrocytes and their myelin sheaths in the context of AD, shedding light on their complex interactions within the disease pathology.

AD is the most common form of dementia and one of the major health challenges of the 21st century¹. With a rapidly aging global population, the number of people with AD is expected to reach about 140 million by 2050, imposing substantial economic and disease burdens on societies and families². According to its initial pathological definition, the disease is characterized by the presence of neurofibrillary tangles within the perikaryal cytoplasm of neurons as well as extracellular amyloid plaques surrounded by reactive glia and abnormally configured neuronal processes. The amyloid hypothesis has served as a powerful theoretical concept, explaining how amyloid peptides trigger a pathological cascade that results in neurodegeneration and dementia³. With advances made in biomarker research, it is now possible to visualize the defining pathological hallmarks of AD longitudinally in patients over time⁴. One of the most important insights from studies on the temporal evolution of disease biomarkers is the definition of the preclinical phase, starting 15–20 years before clinical manifestations with amyloid deposition^{4,5}. A crucial question revolves around understanding the transition from preclinical AD to a state where pathology accelerates, leading to the manifestation of dementia. This transition does not adhere to a simple linear causal dynamic,

where the gradual accumulation of amyloid aligns with progressive neurodegeneration^{4,6–8}. Instead, it is now understood that there exists an extended silent incubation phase during which complex compensatory cellular mechanisms effectively manage the escalating amyloid deposition (Fig. 1). Hence, the brain demonstrates remarkable resilience in managing alterations in amyloid- β (A β) metabolism and deposition.

A major conceptual advancement over the past decade has involved a shift in perspective from a neurocentric cellular view to one that recognizes the contributions of the vasculature and glial cells⁹, including oligodendrocytes, which is underscored by their profound responses toward pathology^{10–12}. Glia possess diverse capabilities to support neurons, and the progression of disease may accelerate when these adaptive functions are compromised, leading to the initiation of detrimental pathological cellular reactions. Although previous reviews have extensively covered the roles of microglia and astrocytes in this process, literature on oligodendrocytes is more limited. Oligodendrocytes are integral to the proper functioning of neuronal circuits, contributing to efficient signal transmission, neuronal health, plasticity and the structural integrity of axons, rendering them key players in many central nervous system diseases. This Review focuses

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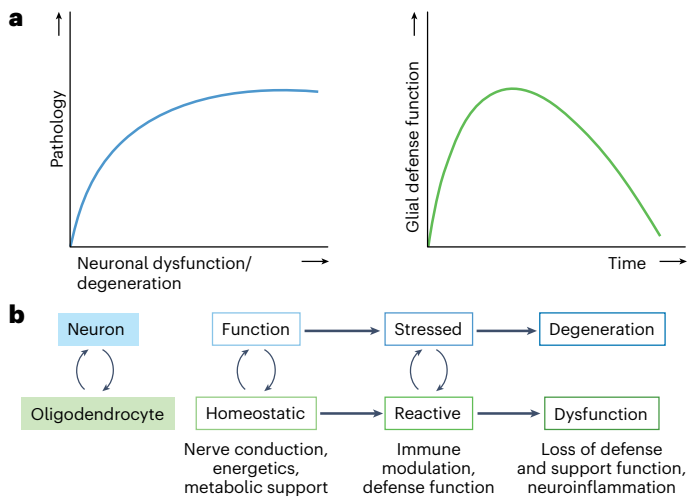


Fig. 1 | Oligodendrocyte response at different stages of AD pathology.

a, Neuronal dysfunction and degeneration occurs with nonlinear kinetics.

Initially, there is compensation and protection, but, as the pathology progresses, the dysfunction becomes increasingly evident. We hypothesize that neurodegeneration is slowed down by glial defense function. Once these protective mechanisms decline, neurodegeneration accelerates.

b, Oligodendrocytes have an intimate anatomical and functional relationship with neurons. The diagram shows the relationship between oligodendrocytes and neurons at different stages of AD pathology. The normal function of oligodendrocytes is to support saltatory nerve conduction, to minimize neuronal energy consumption and to provide metabolic support. Once neurons become affected by increasing pathology, oligodendrocytes may transition into disease-associated states that provide immune modulatory function. The exhaustion of these functions may contribute to neurodegeneration.

on the contribution of oligodendrocytes and their myelin sheaths to the pathophysiology of AD.

Oligodendrocyte pathology in AD

Functions of oligodendrocytes

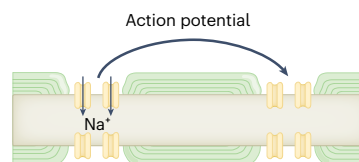
The oligodendroglial lineage can be grouped into oligodendrocyte progenitor cells (OPCs), premyelinating oligodendrocytes and myelinating oligodendrocytes¹³. Single-cell RNA-sequencing analyses of cells from the oligodendrocyte lineage in mice have identified a continuum of multiple different subpopulations or cellular states, ranging from OPCs to immature and mature oligodendrocytes^{14,15}. The morphological classifications of oligodendrocytes trace back to the histological stainings of Pío del Río Hortega, enabling the distinction of various subtypes of oligodendrocytes. These types are largely defined by their microenvironment and the brain region they are located in, including the less-studied perineuronal and perivascular oligodendrocytes. Within the cortex, oligodendrocytes exhibit a highly branched structure, forming up to around 50 different myelin segments on small-diameter axons¹⁶. In most white matter regions, axons are fully covered with myelin with only small segments at the nodes of Ranvier lacking myelin, which forms the structural basis for saltatory nerve conduction. By contrast, gray matter often contains axons that are partially myelinated, featuring segments of myelin interspersed with long unmyelinated stretches, and some axons completely lack myelin altogether. These unmyelinated or partially myelinated axons present opportunities for myelination over the course of life. Such dynamic capability of oligodendrocytes for lifelong myelination supports structural plasticity of the nervous system^{17–19}.

A distinctive characteristic of oligodendroglia is their close anatomical and functional relationship with neurons²⁰ (Fig. 2). OPCs establish direct synaptic contacts with axons, acting as postsynaptic targets

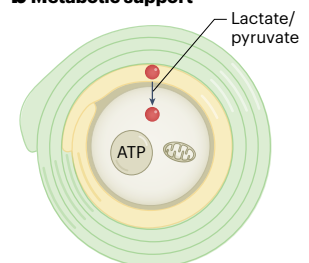
that facilitate the transient activation of ion channel receptors in their processes²¹. Activation of these receptors link neural activity with both proliferation and differentiation of OPCs into mature oligodendrocytes capable of myelination^{22–31}. This process also influences the length and thickness of existing myelin internodes^{23,32}. Therefore, myelin plasticity serves as a mechanism through which experiences and learning can potentially reshape neuronal connections by altering the transmission timing of signals within circuits^{33–35}.

Myelinating oligodendrocytes are metabolically active, maintaining functional connectivity with the underlying axon³⁶. This occurs by a system of noncompact, cytoplasmic-rich regions within myelin, which harbors a transport pathway that facilitates the movement of metabolites and macromolecules to and from the internodal periaxonal space beneath the myelin sheath³⁷. One essential function of oligodendrocytes is to deliver glycolysis products (pyruvate or lactate) to axons using monocarboxylate transporter 1 (MCT1) as a transporter for glycolytic support^{38,39}. This metabolic support must be carefully balanced with axonal energy needs to prevent local acidosis and damage^{40,41}. In addition, oligodendrocytes provide an anti-oxidative defense system for neurons by buffering and chelating accumulating iron⁴². Thus, oligodendrocytes are instrumental for maintaining neuronal homeostasis on many different levels, a process that requires a rich array of sensors to detect deviations in the concentration of various molecules and to respond to such changes, for example, in oxygen, sugars, lipids and ion concentrations. When perceived variations in homeostatic variables occur, these are translated into specific responses using various effector mechanisms, with the goal of restoring homeostatic set points. Hence, neurons are not only structurally connected to oligodendrocytes but also form multiple homeostatic circuits together with oligodendrocytes. It is therefore not surprising that oligodendrocytes show profound changes in neurodegenerative diseases (Fig. 1).

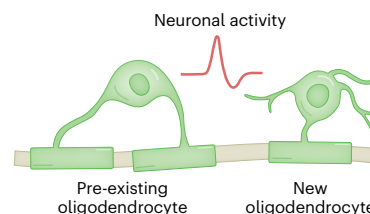
a Saltatory conduction



b Metabolic support



c Myelin plasticity



d Anti-oxidant defense function

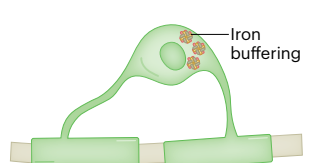


Fig. 2 | Functions of oligodendrocytes. **a**, Saltatory conduction of action potential from one node of Ranvier to the next highly accelerates the propagation of electrical impulses along axons. **b**, Myelinating oligodendrocytes provide metabolic and trophic support to underlying axons through a system of noncompact, cytoplasmic-rich regions known as myelin channels. These myelin channels facilitate movement of metabolites including glycolysis products such as lactate to meet axonal energy requirements. **c**, Neuronal activity stimulates OPCs to differentiate and myelinate in the adult brain. This dynamic capability of oligodendrocytes offers structural plasticity by allowing experiences and learning to reshape existing brain circuits. **d**, Oligodendrocytes provide neurons an anti-oxidative defense system by protecting them against iron-mediated toxicity. Figure created using BioRender.com.

Disease-associated oligodendrocytes in AD

Upon injury, oligodendrocytes transform into a so-called disease-associated state^{15,43–46}. These cells were initially identified in mouse models of amyloidosis using single-cell transcriptomic profiling, where they were found to emerge after the formation of disease-associated microglia and the buildup of amyloid plaques⁴⁶. Similar to disease-associated microglia, the disease-associated oligodendrocyte state is not specific to one type of brain pathology but is observed across various models of brain injury and disease. They appear in experimental autoimmune encephalomyelitis, an experimental model of multiple sclerosis, where they emerge during the priming phase, reach their maximum at the peak phase and gradually diminish during remission^{43,44}. Additionally, they are readily detected in both the cuprizone and lysoclethrin models of remyelination, persisting long into the remyelination phase⁴⁴. In contrast to disease-associated microglia, disease-associated oligodendrocytes are also found in relatively large numbers in mouse models of tau pathology. Even during aging, a relatively large fraction of oligodendrocytes, in particular, in white matter, transition into a disease-associated oligodendrocyte state⁴⁵. Thus, the presence of disease-associated oligodendrocytes reflects a reactive response to brain damage and aging in general (Box 1).

Pathway analysis of differentially expressed genes in disease-associated oligodendrocytes reveals a significant upregulation of immune-related genes, particularly *Serpina3n*, which encodes a serine protease inhibitor associated with immune proteases, as well as the complement component encoded by *C4b*. The function of these two key signature genes in disease-associated oligodendrocytes is unknown, but protective and detrimental functions have been proposed. The product of *Serpina3n* has been demonstrated to inhibit granzyme B, protecting cells from cytotoxic death induced by CD8⁺ T cells⁴⁷. However, it has also been implicated in promoting amyloid plaques in vitro⁴⁶. Likewise, components from the complement system can assist through opsonization-mediated clearing of apoptotic debris and cellular fragments during brain injury, but they can also contribute to neurodegeneration by their role in synapse elimination⁴⁸. There are additional upregulated pathways in disease-associated oligodendrocytes involved in immune signaling, including genes such as *Tnfrsf1a*, *Il1b*, *Il33*, *Hmox1* and *Tnf* and several major histocompatibility complex I genes (*H2-D1*, *H2-K1* and *B2m*). Again, such pathways can provide defense, but they can also amplify the pathology. OPCs can also produce a diverse array of cytokines, chemokines and various receptors for immune-related molecules. For example, in experimental autoimmune encephalomyelitis, OPCs exhibiting an immune transcriptional profile have been demonstrated to acquire phagocytic abilities, present antigens via major histocompatibility complex II and activate CD4⁺ T cells^{15,49}. Two additional, much smaller populations of disease-associated oligodendrocyte states have been identified⁴⁴. One is characterized by the upregulation of genes associated with cell cycle arrest, cell stress and cell survival, such as those involved in the eukaryotic initiation factor 2 (EIF2) signaling pathway and the phosphoinositide-3-kinase (PI3K)–AKT signaling pathway⁴⁴. Another state is defined by the upregulation of interferon-responsive genes and those involved in antigen processing and presentation⁴⁴. All three disease-associated clusters downregulate cholesterol biosynthetic pathways, indicating a shift from their normal function in maintaining myelin sheaths to roles involved in immune modulation, cell stress and survival^{15,43–46}.

Possible functions of disease-associated oligodendrocytes in AD

Microglia and astrocyte reactivity are key features of AD pathophysiology, yet the responses and roles of oligodendrocytes in relation to AD-associated pathology are poorly understood. Although pathway analyses of the disease-associated oligodendrocyte state point to immune modulatory functions, our understanding of what these cells

BOX 1

Oligodendrocyte states in AD

Oligodendrocytes have a crucial role in supporting neurons metabolically. They facilitate the delivery of glycolysis products to axons, thereby fulfilling the energy demands of neurons. Another key function of myelin is to minimize neuronal energy consumption by providing the structural foundation for neurons to conduct by saltatory nerve impulses. Additionally, oligodendrocytes serve as an anti-oxidative defense system for neurons by managing and regulating the accumulation of iron. In AD, a substantial proportion of oligodendrocytes undergo a transition to a disease-associated state characterized by the upregulation of immune-related pathways. This shift in gene expression suggests a change in their functional priorities. Normal homeostatic functions and those related to myelin are downregulated, whereas those essential for defending and supporting neurons against pathological changes are upregulated. Inflammatory signals take precedence because they have higher physiological priority as they coordinate the protective responses to pathology. However, as pathology progresses, oligodendrocytes may lose some of these protective responses and, over time, transition toward exhausted or dysfunctional states, potentially acquiring toxic functions. For example, disrupted myelin sheaths that have lost their ability to metabolically support the axon are known to drive neurodegeneration. Another example is dysfunctional and dying oligodendrocytes that trigger neuroinflammatory cascades. Such damage can be self-perpetuating, as increasing myelin pathology induces greater inflammation, and each subsequent cycle elicits an even greater response.

do in the diseased brain remains limited. One striking feature is the overlap in the transcriptional profile of disease-associated oligodendrocytes and astrocytes⁵⁰, pointing to shared functions of these two cells in response to damage. This suggests that the damage response might occur in a graded manner, with microglia being the first responders, followed by astrocytes and later oligodendrocytes. Together they execute a pan-glial injury response that remains remarkably consistent across various disease models, possibly providing defense and resistance to neuropathological changes. Thus, disease-associated oligodendrocytes may act as a defense system crucial for protecting neurons against pathological changes.

However, as oligodendrocytes adopt a reactive phenotype, they might compromise some of their homeostatic functions. This transformation could potentially affect myelin and oligodendrocytes themselves, thereby altering neuronal network behavior and disrupting the metabolic functions necessary for neuron health. Throughout this process, glia may undergo further conversions as the disease advances, potentially transitioning from protective toward exhausted or dysfunctional states. For example, OPCs within close proximity of amyloid plaques have been shown to become senescent, exhibiting a pro-inflammatory phenotype⁵¹. To unravel the functional role and the dynamics of disease-associated oligodendrocytes in AD models, it will be critical to pinpoint the signaling pathways and transcription factors that trigger their formation. In the case of microglia, this understanding has been transformative, with the identification of triggering receptor expressed on myeloid cells 2 (TREM2) as the key signaling receptor required for the generation of disease-associated microglia^{52,53}. Microglia lacking TREM2 lose their capacity for phagocytosis, lysosomal function and lipid metabolism, all essential functions for defending against AD-related pathology⁵⁴. The formation of disease-associated

oligodendrocytes does not depend on TREM2, suggesting the involvement of signals from other cell types or alternatively activated microglial states driving their development^{44,46}. Motif enrichment analysis for differentially regulated genes in disease-associated oligodendrocytes highlighted transcription factors within the nuclear factor κ B (NF- κ B), signal transducer and activator of transcription (STAT) and interferon regulatory factor (IRF) families⁴³. Another study indicated that abnormal extracellular signal-regulated kinase (ERK)1 and ERK2 signaling was associated with the activation of disease-associated oligodendrocytes⁵⁵. An additional essential question is whether the disease-associated oligodendrocyte states found in mice also manifest in humans. Through immunohistochemistry, OLIG2⁺SERPINA3N⁺ oligodendrocytes have been identified in areas enriched with amyloid plaques in autopsy samples from patients with AD⁴³. However, on a global transcript level, using single-nuclear RNA sequencing, the oligodendrocyte states observed in human brains with AD differ substantially from those seen in mice^{44,46}. Hence, akin to the disease-associated microglial state, which is challenging to capture in autopsy samples from humans with AD, further analyses will be necessary to define the human counterparts in the diseased brain.

Genetic connection of oligodendrocytes to AD

Cell-autonomous AD gene functions in oligodendrocytes

The heritability of AD is substantial, estimated to be at least 50%, and is even higher in early-onset AD^{56,57}, with a small fraction following an autosomal dominant inheritance pattern typically caused by rare variants in genes encoding the A β precursor protein (APP), presenilin 1 (*PSEN1*) or presenilin 2 (*PSEN2*)⁵⁸. By contrast, the *APOE* gene, specifically the *APOE4* allele, substantially contributes to the population's overall risk of AD, with *APOE2* providing protection^{59,60}. The mutations causing early-onset AD induce modifications in the processing of APP, thereby elevating overall A β production or enhancing the relative generation of longer A β peptides, particularly the aggregation-prone A β ₄₂ (ref. 61). Expression of genes in the amyloid-processing pathway (*APP*, *BACE1*, *PSEN1* and *PSEN2*) is highly enriched in oligodendrocytes. Particularly, transcripts for *BACE1* and *PSEN1* rank highest in oligodendrocytes compared to other cell types in mouse and human RNA-sequencing datasets^{62,63}. Additionally, *APP* transcripts are expressed at the highest levels in oligodendrocytes versus other cell types. The elevated expression of both substrate and proteases in the amyloid-processing pathway is also observed at the protein level⁶⁴. Recently, analysis of fresh human brain biopsy samples led to the identification of cellular perturbations in the earliest stages of AD⁶⁵. This was achieved by examining the enrichment of a predefined set of 45 genes responsible for regulating A β production and secretion in each cell type. Intriguingly, the analysis revealed significant positive enrichment for the amyloid gene set not only in excitatory neurons but also in oligodendrocytes compared to other cell types. Moreover, using induced pluripotent stem cell-derived stem cells to experimentally assess the A β -forming potential of these two cell populations revealed that the total amount of A β and the relative ratio of longer A β species was not significantly different between these cells, suggesting that oligodendrocyte-derived A β peptides might contribute to amyloid production. Indeed, using an AD mouse model expressing a humanized and mutated *APP* gene under its own regulatory elements, oligodendrocytes were identified as contributors to A β load^{66,67}. Deleting *BACE1* specifically in oligodendrocytes resulted in a reduction of total amyloid plaque load by approximately 30%, including the soluble fraction of A β peptides^{66,67}. The oligodendrocyte-derived contribution to A β was most pronounced in regions where oligodendrocytes predominate, such as in white matter, where they contributed to the generation of up to approximately 50% of soluble A β peptides. Such accumulation of A β in white matter can be detrimental to oligodendrocytes and myelin, potentially leading to their transition into reactive states. It will be interesting to understand whether the transition into the disease-associated oligodendrocyte state affects the production

of A β . Despite the substantial role of oligodendrocytes in A β generation, the deletion of *BACE1* specifically in neurons was nearly sufficient to abolish amyloid plaque formation, underscoring the essential role of neurons in the generation of A β amyloid plaques. Future work will need to address the spatial and temporal evolution of A β amyloid plaque deposition by neurons and oligodendrocytes. New technologies such as spatial transcriptomics have emerged that enable such high-resolution gene expression analyses within their native spatial context of the AD brain. Applying this technology revealed transcriptional changes within oligodendrocytes in the immediate vicinity of amyloid plaques¹² and unveiled a dynamic response in oligodendrocytes, influenced by the gradual accumulation of amyloid. Unlike the relatively uniform responses observed in microglia and astrocytes, the reaction of oligodendrocytes varies across different brain regions and with respect to increasing amyloid pathology. Initially, the oligodendrocyte response module is upregulated but becomes downregulated in microenvironments with the highest A β accumulation. Determining whether this upregulation during the early phase of amyloid plaque deposition is linked to its role in contributing to A β plaque biogenesis or whether it represents an early reactive response to amyloid remains a subject for further investigation.

Non-cell-autonomous genetic effects of AD risk genes on oligodendrocytes

Non-cell-autonomous effects of AD risk genes on oligodendrocytes are also feasible. One example is *TREM2*, where rare loss-of-function genetic variants have been identified in the coding region of a gene exclusively expressed in microglia, thereby tripling the risk of AD^{68,69}. The considerable effect size of this genetic variation has inspired subsequent functional follow-up analyses to explore the role of *Trem2* in mouse models of AD. This, in turn, yielded compelling evidence supporting the pivotal role of TREM2 in modulating microglial reactivity toward amyloid^{70,71}. In humans, TREM2-expressing microglia are not only important for their reactivity toward amyloid but are also essential for maintaining oligodendrocyte function. Loss of TREM2 function causes Nasu–Hakola disease, also known as polycystic lipomembranous osteodysplasia with sclerosing leukoencephalopathy, a rare autosomal recessive disorder characterized by progressive presenile dementia with white matter degeneration, exhibiting demyelination and accompanied by axon loss and gliosis⁷². There are many other examples of how microglia support oligodendrocyte and myelin integrity^{73–76}. Therefore, it is conceivable that a loss of such function in AD can result in oligodendrocyte dysfunction. Another example is the role of *APOE4* in oligodendrocytes^{77,78}. *APOE* is a gene that is strongly expressed in astrocytes and stands out as one of the most upregulated transcripts in microglia upon activation and transition to their disease-associated state. Although *APOE* expression does occur in neurons and oligodendrocytes, it is notably much less pronounced in these cell types. Yet, the metabolism of oligodendrocytes is intricately linked to APOE, the primary lipoprotein carrier of cholesterol in the central nervous system. The biogenesis of myelin with its multiple layers of lipid-rich membranes enveloping axons requires substantial amounts of cholesterol¹³. Although the majority of cholesterol is synthesized by oligodendrocytes themselves, the deletion of crucial enzymes in the sterol synthesis pathway has revealed that oligodendrocytes can still generate myelin in their absence, albeit at a significantly reduced rate⁷⁹. This suggests that external sources, such as those provided by APOE lipoprotein particles, have a contributing role⁸⁰. Single-nucleus transcriptional profiling of postmortem human brains from *APOE4* carriers, compared with noncarriers, revealed that, among individuals with *APOE4*, numerous biological pathways were observed to be affected across various cell types, including oligodendrocytes⁷⁷. Intriguingly, dysregulation of a group of genes associated with cholesterol biosynthesis and transport were identified specifically in oligodendrocytes. *APOE4* carriers showed lipid droplet formation and cholesteryl ester accumulation in

the cytoplasm of oligodendrocytes. Gene transcripts linked to myelination were downregulated in *APOE4* carriers, even those without AD symptoms, indicating potential early oligodendrocyte dysfunction. Treatment with cyclodextrin, a drug used to extract cholesterol from cell membranes, reduced cholesterol accumulation, increased myelin production and improved learning and memory in *ApoE4*-expressing mice. This study concurs with previous magnetic resonance imaging (MRI) measurements of myelin showing that *APOE4* mediates myelin breakdown in humans⁸¹. Positron emission tomography studies previously demonstrated that *APOE4* allele carriers exhibit reduced cerebral metabolic rates of glucose in adolescence in brain regions susceptible to AD long before potential dementia onset^{82,83}. Together, these studies highlight the role of *APOE4* in oligodendrocyte dysfunction, potentially affecting myelin and leading to impaired nerve conduction, altered neuronal network behavior and disrupted metabolic functions for neurons.

Myelin pathology in AD

Multiple MRI studies have provided evidence of white matter changes in AD, including an overall volume reduction and alterations in microstructure^{84–91}. For instance, using quantitative MRI measures of myelin content alongside cerebrospinal fluid biomarker analyses of amyloid and tau revealed changes in MRI myelin measures during the preclinical phase of AD^{81,87,88}. Additionally, signal abnormalities visualized on T_2 -weighted MRI, known as white matter hyperintensities, predict the onset of AD and are linked to the rate of cognitive decline in AD^{89,90}. Furthermore, within the Dominantly Inherited Alzheimer Network study, it was observed that the volume of white matter hyperintensities is increased in individuals with autosomal dominant, fully penetrant mutations for AD up to two decades before the expected onset of symptoms⁹¹. Postmortem studies suggest that some of the white matter hyperintensities observed in mutation carriers could be linked to cerebral amyloid angiopathy⁹². There is also neuropathological evidence supporting white matter alterations in AD, showing partial loss of myelin and axons, accompanied by reactive astrocytic gliosis, sparsely distributed myeloid cells and hyaline fibrosis of vessels⁹³. Moreover, a decrease in the number of myelinating oligodendrocytes has been observed in postmortem white matter of individuals with AD, although not consistently across all studies^{94–96}. These imaging and neuropathological studies emphasize the crucial role of myelin abnormalities as a fundamental characteristic of AD. One limitation of these human studies is that the majority of dementias in the older population exhibit a mixed etiology involving both vascular and AD-related changes. Consequently, it remains unclear to what extent these changes are influenced by environmental vascular risk factors or genetic risk factors, such as *APOE4*, which is associated with arteriosclerosis and cardiovascular disease. The question arises as to whether the pathology related to amyloid and tau can independently cause damage to oligodendrocytes and myelin. Evidence from animal models of AD suggests that this is indeed the case. Myelin abnormalities have been detected in several different mouse models of amyloidosis and tau pathology^{95,97,98} (Fig. 3). These studies consistently show focal areas of demyelination around amyloid plaques and disruption of myelinated fibers within white matter. Together, these studies raise two important questions: what is the cause of myelin damage? And what are the functional consequences?

The cause of myelin damage in AD

It is important to recall that white matter degeneration with focal areas of myelin pathology is a normal feature of brain aging. In humans, white matter volume reaches its maximum around 40–50 years of age, after which it consistently decreases^{99,100}. Aging of white matter is linked not only to tissue shrinkage but also often to focal lesions visible on MRI as white matter hyperintensities^{101,102}. Electron microscopy studies have revealed that substantial pathological changes occurring during aging are primarily observed in white matter^{103,104}. These changes include

alterations in myelin ultrastructure, such as myelin outfolding, splitting and the accumulation of multilamellar fragments. Deep white matter areas are especially prone to reductions in blood flow and oxygenation as they are located in the most distal part of the arterial circulation. Furthermore, some white matter regions are located in watershed zones (border areas between the anterior and middle cerebral arteries as well as between the middle and posterior cerebral arteries) where blood supply is reduced. These anatomical characteristics may elucidate how vascular changes associated with aging contribute to the heightened susceptibility of aged white matter to hypoperfusion. In addition, cerebral amyloid angiopathy, in which $A\beta$ accumulates in the vessel wall, may contribute to vascular dysfunction in white matter. Oligodendrocytes and myelin are particularly vulnerable to hypoxia-induced oxidative stress¹⁰⁵. One reason is that oligodendrocytes possess an extremely expanded surface area but have restricted capabilities to sustain homeostatic control over their numerous distinct myelin sheaths. Myelin membrane components are embedded within compacted myelin sheaths that undergo very little turnover and therefore are prone to accumulate age-related aging-induced oxidation damage^{106,107}. Interestingly, research in mice has revealed that age-related alterations in oligodendrocytes and myelin occur much earlier in models of AD. Wallerian-like degeneration of axons might explain the accelerated pathology in white matter in AD, but there are also plausible pathological pathways within white matter, such as local effects of soluble and toxic $A\beta$ peptides or oligomers. Indeed, analyses of human AD samples reveal that white matter regions are among the first to display a substantial accumulation of oligomeric $A\beta$ ¹⁰⁸.

The consequences of myelin damage in AD

The functional consequences of an increasing degree of oligodendrocyte and myelin pathology can be categorized into loss of functions and gain of toxic functions (Fig. 4). The functional impairments associated with the loss of oligodendrocyte and myelin functions are well documented in demyelinating diseases. These disruptions encompass aspects such as nerve conduction, neuronal network behavior and metabolic and other supportive functions for neurons. Intriguingly, enhancing myelin renewal with the promyelinating drug clemastine has been shown to reverse cognitive dysfunction in mouse models of AD⁹⁸.

Chronic alterations to oligodendrocytes and myelin may also cause gain of toxic functions. One example is the link between myelin dysfunction and amyloid plaque deposition in models of AD. Using mouse mutants and toxins associated with myelin dysfunction or demyelination, a recent study found that myelin breakdown drives amyloidosis. Myelin dysfunction was found to result in buildup of the $A\beta$ -producing machinery within axonal swellings, leading to an increased release of $A\beta$ peptides¹⁰⁹. Additionally, despite an overall increase in their numbers, plaque-associated microglia were reduced. Intriguingly, these disease-associated microglia, which are typically responsible for clearing amyloid plaques, appear to be drawn to areas of myelin damage. This observation implies that microglial dysfunction could be another potential mechanism contributing to the increased amyloid plaque load in AD mice with impaired myelin function¹¹⁰.

Another way in which gain of toxic functions can occur is through dysfunctional oligodendrocytes. Studies on models of demyelination and dysmyelination have provided evidence indicating that progressive axon degeneration predominantly occurs in axons where disturbed myelin sheath persists^{111,112}. Dysfunctional myelin may not only lose its ability to metabolically support the axon but also might be unable to provide resistance to injury-related signals, such as reactive oxygen species. The underlying cause may involve a breakdown of noncompact cytoplasmic channels within myelin sheaths, crucial for the metabolic connection between myelin and neurons³⁷.

One more example of how myelin degeneration might drive detrimental pathways is linked to its pro-inflammatory role. Myelin damage can lead to secondary immune responses, as demonstrated in a

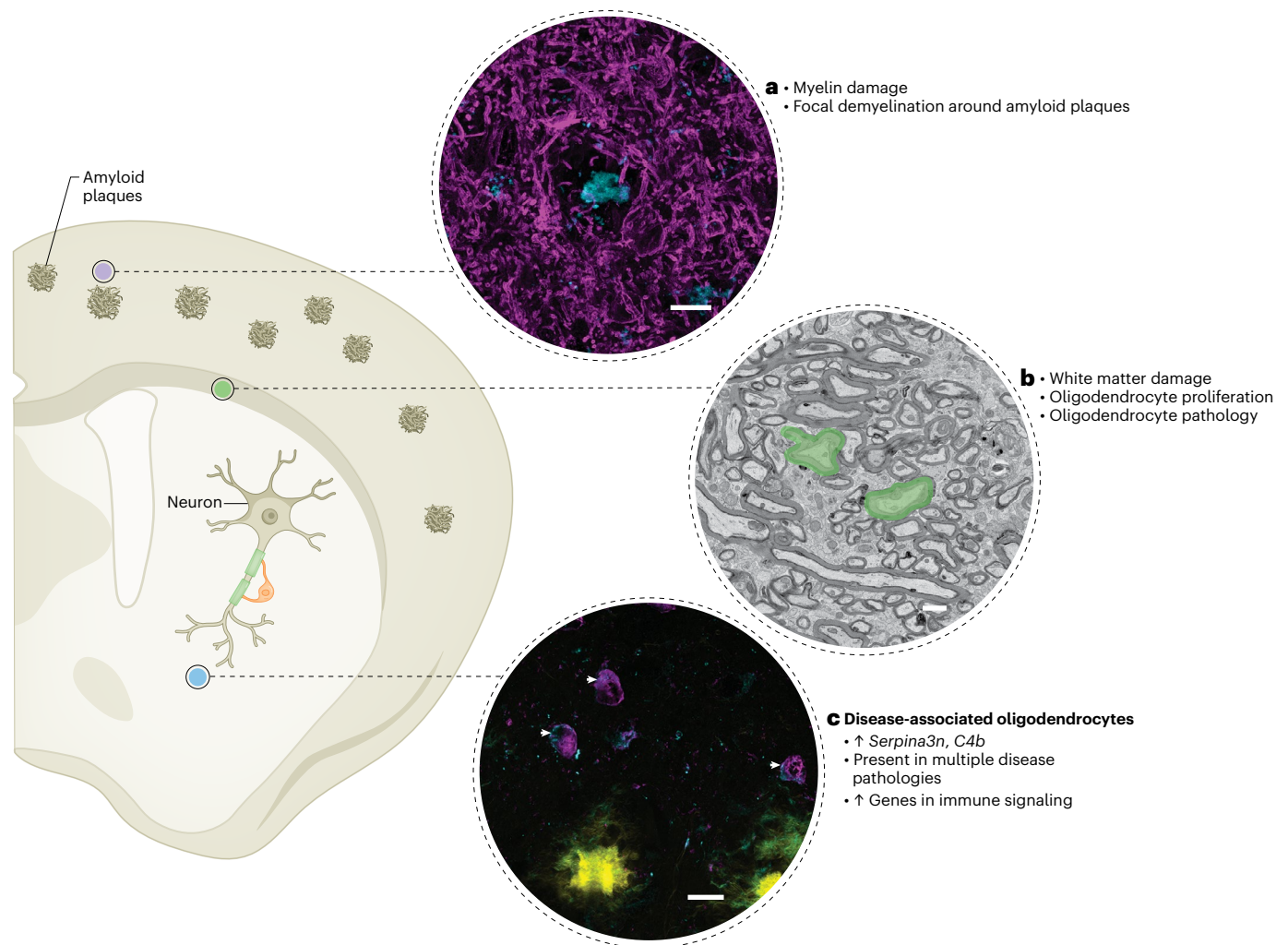


Fig. 3 | Oligodendrocyte and myelin damage in AD. **a, b**, Myelin and oligodendrocyte pathology, including focal areas of demyelination around amyloid plaques, damage to white matter tracts and changes in myelin ultrastructure, has been detected in different AD mouse models. **a**, Myelin basic protein (MBP) (magenta) around amyloid plaques (cyan) in 5xFAD animals aged 10 months. **b**, Scanning electron microscopy image shows abnormal myelin

(green) in the corpus callosum of 5xFAD animals aged 10 months. Scale bars, 10 μ m (**a**) and 1 μ m (**b**). **c**, Oligodendrocytes under pathological conditions transition to a disease-associated oligodendrocyte state, marked by upregulation of *Serpina3n* and *C4b*. Image shows SERPINA3N (cyan)-positive oligodendrocytes (magenta) around amyloid plaques (yellow) in the cortex of 5xFAD animals aged 10 months. Scale bar, 10 μ m. Figure created using [BioRender.com](https://www.biorender.com).

genetically inducible mouse model of acute demyelination, resulting in late-onset chronic demyelination marked by increased numbers of T lymphocytes¹¹³. In line with this concept, a recent study using a mouse model of central nervous system amyloidosis identified a gradual buildup of disease-associated oligodendrocytes and myelin abnormalities along with a small population of CD8⁺ T cells¹¹⁴.

What are the consequences of myelin pathology? Demyelination triggers OPC proliferation and differentiation into mature oligodendrocytes. During this transition, they have to pass a particularly vulnerable premyelinating oligodendrocyte cell state, which is characterized by high biosynthetic activity, increasing susceptibility to endoplasmic reticulum stress and cell death¹⁹. For example, a mouse model for amyotrophic lateral sclerosis showed that the rate of proliferation for OPCs is significantly increased, but newly differentiated oligodendrocytes undergo early cell death¹¹⁵. Also in models of AD, OPC proliferation increases without subsequent transition into fully differentiated oligodendrocytes^{95,116}. Instead, there appears to be an accumulation of premyelinating oligodendrocytes that exhibit severe pathological alterations with abnormal swollen processes¹¹⁴. It is thus conceivable that progressive myelin damage exhibits the characteristics of a self-propelling positive feedback system, in which increasing myelin

pathology induces greater inflammation and each subsequent loop elicits an even more substantial response. Indeed, the depletion of CD8⁺ T cells was sufficient to break this vicious cycle, possibly via the accumulation of abnormally activated microglia, which displayed myelin-damaging activity¹¹⁴.

Link between myelin and tau pathology

Braak et al.¹¹⁷ outlined a specific pattern of tau aggregate distribution starting in brain stem nerve cells, progressing to the transentorhinal region and spreading to the olfactory bulb, the entorhinal region and hippocampal formation. In later stages of AD, tau reaches the basal temporal neocortex and expands further to the temporal, insular and frontal neocortex. Biochemically, tau aggregates evolve from nonfibrillary pretangles in axons to rigid fibrillary threads and neurofibrillary tangles in the neuronal soma and dendrites. This transition occurs gradually, accelerating over time, with initial pretangle stages appearing in adolescence before amyloid plaques are detectable¹¹⁸. The progression from pretangles to neurofibrillary tangles spans a lifetime, although not all pretangles convert. Therefore, a crucial task is to determine the factors that propel the expansion of neurofibrillary tangles across broader areas of the brain. Interestingly,

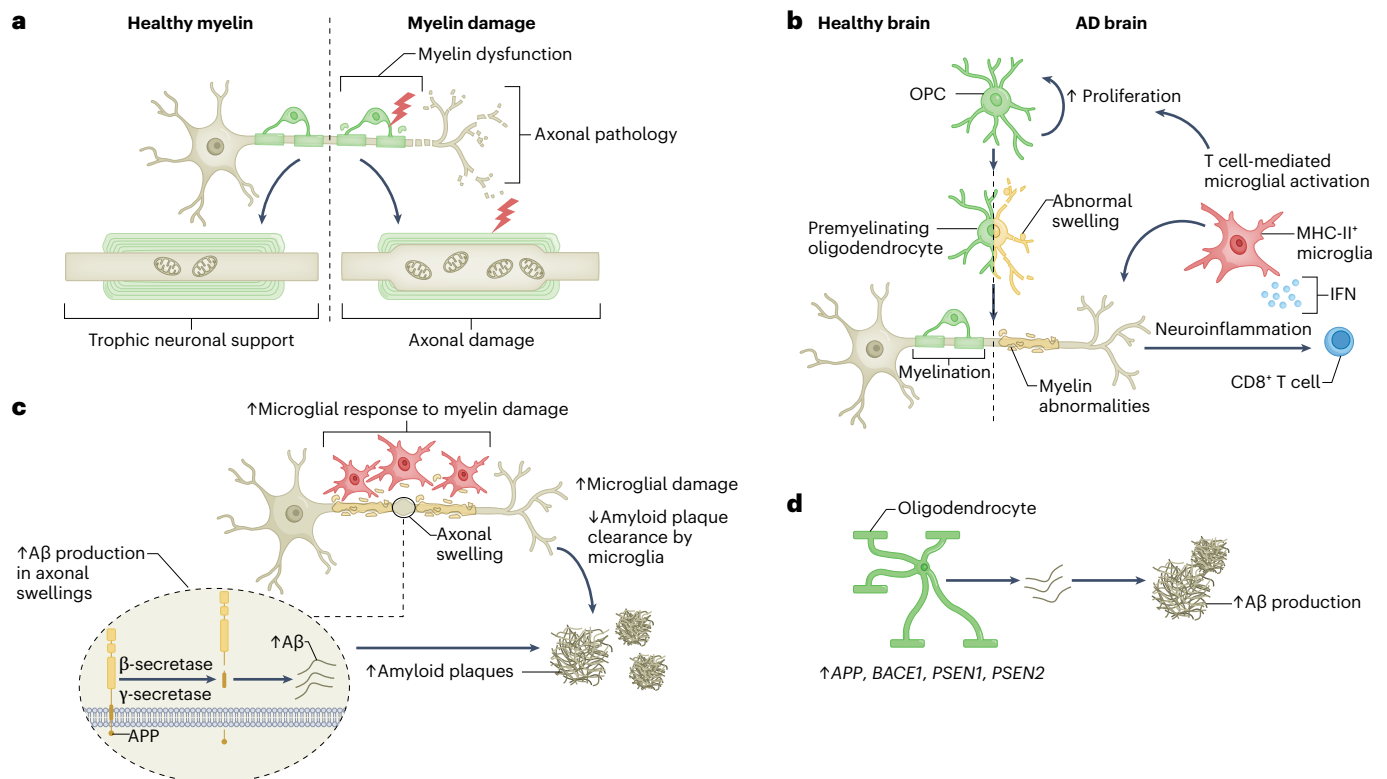


Fig. 4 | Oligodendrocyte response in AD. **a**, Oligodendrocytes provide metabolic and trophic support to neurons. Myelin damage not only disrupts oligodendroglial support mechanisms, but dysfunctional myelin itself induces axonal damage and is detrimental for axonal survival. **b**, Myelin dysfunction drives amyloid plaque deposition in AD. Mechanistically, myelin dysfunction leads to accumulation of Aβ-producing machinery in axonal swellings, leading to higher Aβ production. Additionally, myelin damage engages disease-associated microglia, interfering with their ability to clear amyloid plaques; IFN, interferon;

MHC-II, major histocompatibility complex class II. **c**, Myelin damage in AD is marked by increased CD8⁺ T cell infiltration in the brain. Mechanistically, CD8⁺ T cells abnormally activate microglia to damage myelin, driving a self-propelling loop, in which enhanced myelin damage drives further neuroinflammation. **d**, Oligodendrocytes show enrichment in genes associated with the amyloid-processing pathway (such as *APP*, *BACE1*, *PSEN1* and *PSEN2*), thereby contributing to the total Aβ plaque load. Figure created using BioRender.com.

the development pattern of neurofibrillary tangles inversely mirrors the pattern of myelin formation during development¹¹⁸. Myelination is initiated in regions dedicated to general brain homeostasis within the brain stem, progresses through sensory pathways, advances to motor areas, followed by projection pathways and finally culminates in association fibers. This sequential process moves from caudal to rostral, posterior to anterior and central to peripheral locations. The entire process takes several decades to complete, beginning at a high rate around birth and persisting at a lower level well into adulthood. Despite this prolonged process, it is insufficient to completely eliminate the intermittent pattern of sparsely myelinated axons in the cortex^{17,119}. Hence, the brain regions that undergo myelination later in development generally involve oligodendrocytes that generate isolated myelin sheaths on specific axonal segments. Neurofibrillary tangles predominantly form in cortical projection neurons with long and poorly myelinated axons. It is still unclear whether a connection exists between late-maturing oligodendrocytes forming sparsely myelinated axons and the susceptibility of these neurons to tau pathology. Previously, it has been proposed that late-myelinating oligodendrocytes are particularly vulnerable to noxious insults such as oxidative stress¹⁰⁵. Loss of these cells may have circuit-level consequences with changes in overall neuronal network behavior. For example, impairment of intermittent myelination, frequently detected in parvalbumin-positive basket interneurons, may lead to alterations in neuronal firing patterns¹²⁰. How such alterations in oligodendrocytes and their pattern of myelination in the cortex promote tau aggregation and/or spreading is not clear.

There is an alternative explanation for why neurofibrillary tangles start to form in poorly and late-myelinated areas. The complexity of intracortical circuits shows an inverse correlation with myelin content (Fig. 5). Regions with lower myelin content exhibit more complex intracortical circuits characterized by neurons with large dendritic arbors and numerous synapses¹²¹. These areas appear to be highly metabolically active, demonstrating higher rates of aerobic glycolysis than regions of higher myelin content. One factor responsible for ongoing aerobic glycolysis in the brain is the need to support membrane-bound, ATP-dependent processes essential for action potential generation and synaptic transmission. Additional processes crucial for neuronal transmission, such as axonal transport and neurotransmitter synthesis, contribute greatly to the heightened energy requirements. A key function of myelin is to reduce neuronal energy expenditure. To generate energy, neurons operate at a high rate of oxidative metabolism, posing the risk of generating reactive oxygen species such as superoxide anions and hydrogen peroxide, byproducts of oxidative phosphorylation. Therefore, myelin may indirectly protect neurons from age-associated oxidative stress, potentially contributing to the prevention of tau aggregation and spreading. Consistent with this, an imaging study using MRI measures for myelin and tau positron emission tomography in individuals with AD concluded that higher levels of myelin are associated with lower susceptibility to the spreading of tau pathology¹²². This raises the question of whether promoting adaptive myelination during periods when the brain is still capable of adult myelination offers protection against tau spreading later on^{35,123,124}. Studies in mice and humans have shown that measures of

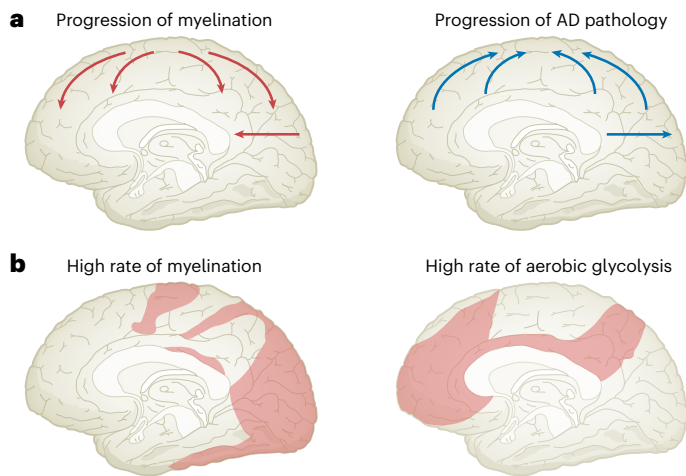


Fig. 5 | Link between myelination and AD pathology progression. **a**, Left, myelination begins in the brain stem, progressing to sensory and motor fields, followed by association fibers. This results in dense myelination in regions undergoing myelination first, in comparison to regions myelinated later in development, which often display intermittent sparse myelination patterns. Right, an inverse relationship between progression of AD pathology and myelination. Pathology usually begins in the anterior temporal mesocortex, progressing to the association fibers, followed by sensory and motor fields. **b**, Diagram showing an inverse relationship between regions with high rates of myelination (left) and high rates of aerobic glycolysis (right). Sparsely myelinated brain regions often contain complex intracortical circuits, exhibiting higher rates of metabolism and aerobic glycolysis. Figure created using [BioRender.com](https://www.biorender.com).

myelin structure or patterns change upon learning new tasks¹²⁵. For instance, motor learning promotes the generation of more oligodendrocytes and new myelin sheaths in rodents^{126–128}. Conversely, sensory-motor deprivation or social isolation results in hypomyelination^{129,130}. These studies collectively provide evidence that myelin biogenesis is influenced by experience, raising the question of whether this process could be leveraged to enhance resistance and resilience against tau pathology.

Conclusion

Previous work has revealed glial reactivity as a major contributor to the pathology of AD, although limited attention has been given to oligodendrocytes. Nevertheless, oligodendrocytes react to amyloid and tau pathology, and myelin alterations have been consistently observed in both brains from humans with AD and corresponding mouse models. Identifying the roles of oligodendrocytes and myelin as well as understanding how their functions evolve during pathology is a crucial task for future research. In this context, we propose that oligodendrocytes have a role in providing defenses and resistance against accumulating pathology, particularly when other cells, such as microglia, become overwhelmed. We hypothesize that the initial response of oligodendrocytes is adaptive, initiating an effector program to safeguard neuronal function and viability. However, when these compensatory mechanisms are overwhelmed, maladaptive pathological processes contribute to the progressive decline in neuronal function. Possibly, dysfunction in oligodendrocytes and myelin contributes to self-reinforcing pathological processes, including amyloid formation and tau spreading, ultimately leading to neurodegeneration and dementia. Oligodendrocytes should therefore no longer be seen as passive elements but rather as dynamic facilitators contributing to the progression of the disease. In the years ahead, it is essential to enhance our comprehension of how oligodendrocytes respond and operate at various stages of AD. This knowledge, along with strategies aimed at enhancing oligodendrocyte well-being or increasing myelin, could hold major therapeutic promise for AD.

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Author contributions

M.S. and S.K. conceived the theme, scope and structure of the Review, wrote the manuscript and drafted the figures.

Competing interests

The authors declare no competing interests.

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